PROPOSAL: INTEGRATED CARBON CAPTURE AND MINE SAFETY MONITORING SYSTEMS

1. Executive Summary

This research initiative seeks to revolutionize the current coal mining landscape by integrating cutting-edge Carbon Capture technologies with traditional excavation methods. The primary goal is to reduce the carbon footprint of national mining operations while simultaneously increasing output efficiency through the use of automated data streams.

2. Personnel and Leadership

The project will be spearheaded by the Principal Investigator, Dr. Vikram Seth. Dr. Seth holds a Doctorate in Geo-Technical Engineering and has spent 20 years studying the tectonic shifts in underground mines. He is currently a Senior Fellow at the National Institute of Mining Research.

3. Technical Methodology: Gasification

A core component of this study is Coal Gasification. Unlike traditional combustion, our method converts coal into syngas, which can then be used for Blue Hydrogen production. This process is essential for meeting the 2026 sustainability targets set by the Ministry and ensures that deep-seated coal reserves are not wasted.

4. Mine Safety Monitoring

To protect the workforce, we are implementing a proprietary Mine Safety Monitoring framework. This system utilizes localized sensors to detect seismic vibrations and gas leaks. By focusing on real-time telemetry, we can provide immediate warnings to personnel in the event of structural instability or ventilation failure.

5. Geological Analysis

The research team has developed a new model for Fault Prediction. In deep-shaft mining, identifying geological faults 2m to 5m ahead of the face is critical. Our algorithm analyzes historical data to predict where faults will occur, significantly reducing the risk of unplanned roof collapses.

6. Environmental Impact

In alignment with the Waste to Wealth initiative, the project includes a sub-study on utilizing fly ash for structural brick manufacturing. Furthermore, the plan addresses Pit Lake Management by converting abandoned mining pits into sustainable fish hatcheries and community water reservoirs.

7. Financial Estimates

The financial framework is designed for maximum efficiency. The Total Budget requested for this three-phase operation is Rs. 42,00,000. This includes equipment procurement, site labor costs, and computational overhead for the predictive algorithms.

8. Project Duration

The anticipated Project Duration is 18 months. The first six months will focus on site preparation, followed by eight months of active testing, and a final four-month period for data synthesis and reporting to the Central Mine Planning and Design Institute.

9. Future Scope: Alternative Energy

Future iterations of this project will explore the integration of Perovskite Solar cells to power the underground sensor network. Additionally, we are investigating Blue Hydrogen applications to provide clean fuel for heavy mining machinery on-site.

10. Conclusion

In conclusion, this proposal offers a comprehensive solution to the twin challenges of safety and sustainability. By combining Coal Gasification with advanced safety tech, we are ensuring the long-term viability of the Indian coal sector.